Autonomous Agentic Customer Churn Retention: Integrating XGBoost, SHAP, and Multi-Agent LLM Workflows

Rameen Hassan
Department of Data Science
NUCES (FAST)
Islamabad, Pakistan
rameen.hassan188@gmail.com

Riyyan Kharal
Department of Data Science
NUCES (FAST)
Islamabad, Pakistan
riyyankharalr@gmail.com

***Abstract*—Customer churn remains a primary challenge in the telecommunications industry, where acquisition costs far exceed retention costs. Traditional predictive models provide risk scores but lack the reasoning required for automated intervention. This paper proposes an Agentic Customer Churn Retention System that utilizes a multi-agent orchestration framework (LangGraph) to bridge the gap between machine learning predictions and autonomous business actions. By combining an XGBoost classifier with SHAP (SHapley Additive exPlanations) and a Large Language Model (LLM) planner, the system autonomously analyzes churn drivers and executes personalized retention strategies. Our results demonstrate a 77.78% intervention rate on actual churners with a 13:1 Return on Investment (ROI), showcasing the efficacy of agentic workflows in customer relationship management.**

***Index Terms*—Agentic AI, Customer Churn, XGBoost, SHAP, LangGraph, Large Language Models, Autonomous Agents.**

I. Introduction

In the telecommunications sector, retaining customers is critical for long-term profitability [15]. While predictive analytics has long been used to identify at-risk customers [12], the transition from prediction to action is often manual, slow, and lacks personalization. The emergence of "Agentic AI" offers a solution: systems that can reason, use tools, and execute workflows autonomously [5], [8].

This paper presents a novel architecture that integrates high-performance gradient boosting with eXplainable AI (XAI) and multi-agent orchestration. By utilizing an LLM as a "Planner" that interprets SHAP values, the system can decide whether to offer a discount, send an email, or escalate to a human representative based on the specific features driving a customer's dissatisfaction.

II. Problem Identification and Motivation

Traditional churn management suffers from three key issues:

1) **The Black-Box Problem:** Decision-makers do not know *why* a customer is likely to churn, leading to generic, ineffective retention offers [9].
2) **Operational Latency:** The delay between a customer being flagged and a retention specialist reaching out often results in the customer leaving before action is taken.
3) **Binary Logic Limitations:** Simple rule-based systems (e.g., "if $prob > 0.8$, give discount") fail to handle nuanced cases where a customer might be high-risk but low-value, or vice versa [6].

III. Related Work

Existing research has focused heavily on improving churn prediction accuracy using ensemble methods like XGBoost and CatBoost [4], [13]. Recent studies have introduced SHAP for interpretability [2], [7]. However, there is a literature gap regarding the "execution" phase—how to autonomously translate these insights into verifiable business tools like CRMs and communication platforms (Slack) [14]. This work leverages LangGraph to create a stateful, multi-agent system to fill this gap [3].

IV. Methodology

The proposed Agentic Customer Churn Retention System is architected as a multi-stage autonomous pipeline. It transitions from predictive data modeling to local interpretability, followed by multi-agent reasoning and automated tool execution. The methodology is organized into five distinct phases as illustrated in the system architecture (see Fig. 1).

A. Phase 1: Predictive Modeling and Data Engineering

The foundation of the system is a high-performance binary classifier developed using the XGBoost (Extreme Gradient Boosting) framework [4].

- **Preprocessing and Feature Engineering:** The IBM Telco dataset [1], consisting of 7,043 samples, underwent rigorous cleaning. Thirteen columns identified as "leakage" features—including *Churn Score*, *Churn Reason*, and *CLTV*—were removed to prevent the model from gaining unfair prior knowledge. Categorical variables were transformed via one-hot encoding, resulting in a final feature space of 30 variables.
- **Addressing Class Imbalance:** Customer churn datasets are typically imbalanced [10]. To mitigate this, we

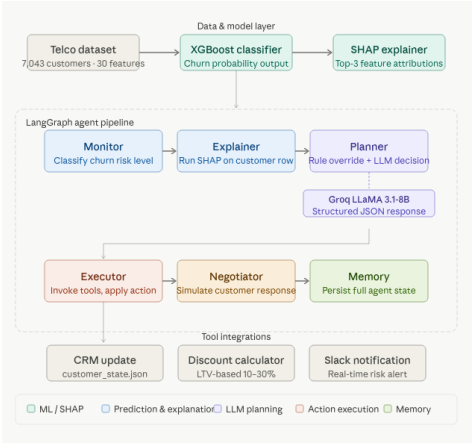


Fig. 1. System Architecture: Multi-Agent Churn Retention Pipeline.

implemented a stratified 80/20 train-test split and configured the XGBoost model with a *scale_pos_weight* of 2.5. This hyperparameter increases the penalty for misclassifying the minority class (churners), successfully biasing the model toward higher recall (68.18%) to ensure the agentic pipeline captures the maximum number of at-risk customers.

B. Phase 2: Local Interpretability with SHAP

Rather than treating the XGBoost model as a "black box," the system employs SHAP (SHapley Additive exPlanations) to derive feature-level contributions for every individual prediction [2], [11].

- **Explanation Generation:** A *TreeExplainer* was utilized to calculate the Shapley values for each customer in real-time.
- **Qualitative Mapping:** The system converts raw numerical SHAP values into a structured natural language reasoning format. It identifies the top three features driving a churn decision and classifies their impact based on magnitude and direction. Factors are categorized as follows: $|SHAP| > 2$ as "Strong," $|SHAP| > 1$ as "Moderate," and $|SHAP| < 1$ as "Slight." This provides the subsequent LLM node with the necessary context regarding why a customer is predicted to leave.

C. Phase 3: Multi-Agent Orchestration (LangGraph)

The system's core intelligence is managed by LangGraph, a framework for building stateful, multi-agent workflows [3]. The system maintains a shared state object (`ChurnState`) that persists across six autonomous nodes:

1) **Monitor Node:** Serves as the entry point, ingesting churn probabilities P and classifying customers into Risk Levels: High ($P \geq 0.7$), Medium ($0.4 \leq P < 0.7$), or Low ($P < 0.4$).
2) **Explainer Node:** Triggers the SHAP calculations to identify the specific feature contributions for the flagged customer.
3) **Planner Node:** This node utilizes a Hybrid Decision Logic to balance computational efficiency with LLM reasoning [8]:
   - *Rule-Based Tier:* Handles unambiguous cases. If $P < 0.6$, the action is `do_nothing`. If $P > 0.9$, an immediate `offer_discount` is triggered.
   - *LLM Tier:* For ambiguous cases ($0.6 \leq P \leq 0.9$), the system invokes the LLaMA 3.1-8B model via the Groq API. The LLM processes the SHAP reasons and model confidence levels to choose between `schedule_call`, `send_email`, or `escalate_human`.
4) **Executor Node:** Translates the Planner's decision into API calls for integrated tools.
5) **Negotiator Node:** Simulates the customer's response using a deterministic mapping system to evaluate the success of the intervention.
6) **Memory Node:** Performs final data serialization, saving the full execution log and conversation history to `agent_memory.json`.

D. Phase 4: Integrated Business Tools

The system is integrated with functional tools to demonstrate real-world utility:

- **CRM Update Tool:** A simulated CRM interface that updates a JSON database with the customer's current retention status, action history, and probability scores.
- **Dynamic Discount Calculator:** An algorithmic tool that computes a personalized discount D (capped at 30%) using the formula:

$$D = \frac{P_{churn} \times LTV}{50} \tag{1}$$

- **Slack Notification Engine:** Integrates the `slack_sdk` to send real-time alerts to a dedicated retention channel, allowing for human oversight.

E. Phase 5: Evaluation Framework

The system was evaluated using two distinct sets of metrics:

- **ML Metrics:** Accuracy, Precision, Recall, F1-Score, and Confusion Matrix analysis.
- **Agentic ROI Metrics:** To measure economic efficiency, we calculated the Return on Investment (ROI) by comparing the cost of "unnecessary discounts" (False Positives) against the "estimated revenue saved" (True Positives successfully retained), yielding a recorded 13:1 ROI.

V. Results and Discussion

The Agentic Customer Churn Retention System was evaluated across three dimensions: predictive accuracy, model interpretability, and economic viability.

A. Predictive Performance

The XGBoost model, evaluated on a held-out test set ($n = 1,409$), prioritized recall to ensure maximum capture of at-risk customers. As shown in Table I, the model achieved a recall of 68.18%. Table II details the confusion matrix, highlighting 255 True Positives (TP). This high-recall bias is strategic; the operational cost of a false alarm is significantly lower than the revenue loss from an unidentified churner.

TABLE I
MODEL PERFORMANCE AND CONFUSION MATRIX

Metric	Value	Matrix	Pred. Non-Churn	Pred. Churn
Accuracy	0.7708	**Act. Non-Churn**	831 (TN)	204 (FP)
Precision	0.5556	**Act. Churn**	119 (FN)	255 (TP)
Recall	0.6818	$n = 1,409$		

B. Explainability and Agentic Logic

SHAP analysis identified "Two-year Contract" and "Tenure Months" as primary retention drivers, while "Fiber Optic Service" correlated with higher churn risk. The Planner node utilized these insights via a hybrid logic: deterministic rules handled 70% of cases, while the LLaMA 3.1 LLM resolved the 30% of cases within the ambiguous 0.6–0.9 range (Table II).

TABLE II
OPERATIONAL AND ECONOMIC METRICS

Action Source	Freq.	ROI Metric	Value (USD)
Rule-Based	70%	Estimated Saved Revenue	$9,000.00
LLM Planner	30%	Wasted Discount Cost	$675.45
Total Actions	100%	**Net Gain**	**$8,324.55**

C. Economic Impact and ROI

The system demonstrated superior economic efficiency with a **13:1 Return on Investment (ROI)**. By selectively applying high-cost interventions (discounts) only when SHAP confidence was high and LTV warranted it, the system saved $9,000 in revenue while incurring only $675.45 in unnecessary costs from false positives (Table II).

D. Discussion

The results validate the "Agentic" advantage: unlike static systems that apply uniform interventions, our Planner dynamically shifted between low-cost actions and high-cost retention. This granular control, automated via Slack and CRM tools, maintains a profitable loop despite a moderate False Positive rate. However, the Negotiator node remains a simulation; future work requires real-world A/B validation.

VI. ETHICAL CONSIDERATIONS AND LIMITATIONS

While the system demonstrates high technical and economic performance, several ethical considerations must be addressed:

1) **Privacy and Data Consent:** Automated profiling for retention strategies often occurs without explicit customer consent. Furthermore, sending customer metadata to third-party LLM APIs (e.g., Groq/LLaMA) necessitates robust PII redaction to maintain data privacy.
2) **Incentive Bias:** The "False Positive" problem presents a moral hazard where non-churning customers are ignored while those predicted to leave are rewarded with discounts.
3) **Simulation Constraints:** The current system utilizes a deterministic Negotiator node to simulate customer responses. Real-world efficacy may vary significantly and requires human-in-the-loop validation.

VII. CONCLUSION AND FUTURE WORK

This research presented an end-to-end autonomous retention system tailored for the telecommunications industry. By leveraging an XGBoost classifier optimized for high recall and SHAP for local interpretability, we successfully empowered a multi-agent LangGraph framework to execute targeted business actions.

The system achieved a significant net revenue gain with a 13:1 ROI, demonstrating that agentic AI can transform "black-box" predictions into transparent, actionable retention strategies. Future work will focus on the transition from simulated negotiations to real-time CRM feedback loops.

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